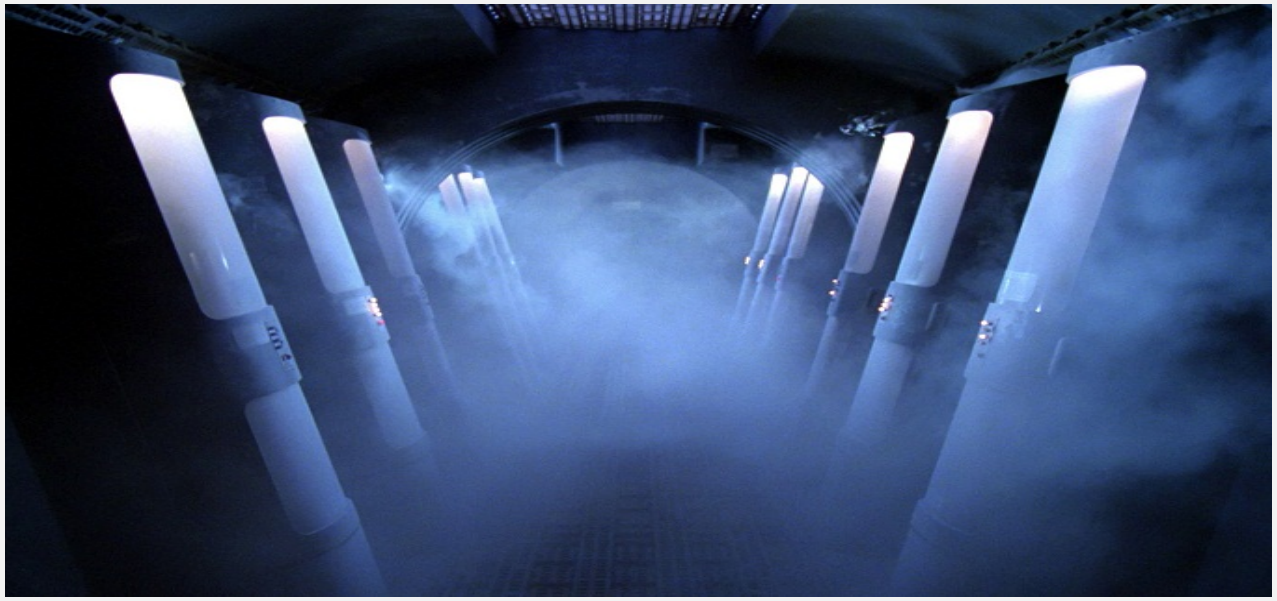
Object Serialization



Object Serialization: *Time to freeze-dry?*

- Serialization is an (automatic) way to save and load the state of an object from a stream.The serialized object could then be stored in a file, database or sent over the network.
- To be able to serialize a class, two conditions must be met:
  - The class must implement the `java.io.Serializable` interface. The interface is empty and serves simply as permission to serialize!
  - All of the fields in the class must be serializable. If a field is not serializable, it must be marked `transient`.
- `ObjectInputStream`, `ObjectOutputStream` are special stream subclasses used to serialize objects. Subclasses of `Serializable` classes are also serializable.
- When an object is serialized, any object references it contains are also serialized. Serialization can capture entire "graphs" of interconnected objects and put them back together on the receiving end.

[Examples are in the repository folder `examples/serialization`.]

Basic Examples

- Example 1: SaveTable.java, LoadTable.java.
  - Shows how to (de)serialize a standard Java data structure.
- Example 2: MioAlma.java, Cryogenics.java
  - Shows how to (de)serialize our own class. Also shows the affect of the `transient` keyword.

Serial Version UID

- By default, serialization uses a calculated hash based on a given source file — method names, field names, field types, access modifiers, etc., and compares that hash value against the hash value in the serialized stream.
- To convince the compiler that two versions of a class are equivalent, each class that is serializable can declare a version number, called a `serialVersionUID`
- The `serialVersionUID` is declared via a field named "serialVersionUID" that must be static, final, and of type long: `<access-modifier> static long serialVersionUID = <UID>L;`
- Use the program `serialver` (bundled with Java) to generate the `serialVersionUID` for a given class. Or use your IDE to generate it for you!

Serialization versus Refactoring

- Serialization permits some variation in classes such that serialized objects can be revived by various versions of a class.
- Using `serialVersionID`, Java serialization can automatically manage the following refactorings:
  - Adding new fields to a class
  - Changing the fields from static to nonstatic
  - Changing the fields from transient to nontransient
- Going the other way requires additional processing depending upon the amount of backwards compatibility desired.

Serialization Experiments

Examples: MioAlma.java, Cryogenics.java, MioAlmaDos.java.

1. Run and freeze a `MioAlma` object (version 1) using the `Cryogenics` program.
2. Refactor the `MioAlma` class to add an email field. Or simply copy and paste the code from MioAlmaDos.java over the code in MioAlama.java to create version 2 of the class.
   - 2.1 Using this new class, attempt to revive an `MioAlma` version 1 serialized object.
   - 2.2 Note that this will fail as the class has changed and we didn't use a serial version UID.
3. Revert `MioAlma` to version 1. Add the serial version UID to the Mio class (version 1)
   - 3.1 Refactor `MioAlma` into version 2 (as before).
   - 3.2 Using this new class, attempt to revive an `MioAlma` version 1 serialized object.
   - 3.3 This time it will work. Note that the an email field will be set to null.
4. Run and freeze a `MioAlma` object (version 2)
   - 4.1 Revert `MioAlma` back to version 1.
   - 4.2 Revive a `MioAlma` version 1 object from a freeze-dried `MioAlma` version 2 object!

SerialVersionUID

- It is strongly recommended that all serializable classes explicitly declare `serialVersionUID` values.
  - *Reason*: The default `serialVersionUID` computation is highly sensitive to class details that may vary depending on compiler implementations, and can thus result in unexpected `InvalidClassException` exceptions.

Custom Deserialization

- Simple deserialization may not be enough to reconstruct the full state of an object. Objects can do their own setup after deserialization by implementing the `readObject()` (as well as `writeObject` method).
- The methods are defined in the interfaces `ObjectInput` and `ObjectOutput` in `java.io` package.

```
private void readObject(ObjectInputStream stream)
        throws IOException, ClassNotFoundException;
private void writeObject(ObjectOutputStream stream)
        throws IOException;
```

- Example:

```
private void readObject (ObjectInputStream s)
{
  s.defaultReadObject(); //standard deserialization
  initialize(); //our custom initialization
  //call optional method after customization
  if (isRunning)
      start();
}
```

Overriding Deserialization

- Use the `java.io.Externalizable` interface to override the serialization process.

`void readExternal(ObjectInput in)`
`void writeExternal(ObjectOutput out)`

- `readExternal`: Implement the method to restore its contents by calling the methods of DataInput for primitive types and readObject for objects, strings and arrays.
- `writeExternal`: Implement the method to save its contents by calling the methods of DataOutput for its primitive values or calling the writeObject method of ObjectOutput for objects, strings, and arrays.

Serialization and Security

- Serialized objects contain all the data in a easily readable format so it isn't secure. The security can be dealt with in multiple ways.
  - Override the `readObject` and `writeObject` and implement our own algorithms for obscuring the data.
  - Encrypt and sign the entire object using `javax.crypto.SealedObject` and/or `java.security.SignedObject` wrapper. However, this requires managing symmetric keys.
  - For secure transport over the network, use SSL (Secure Sockets Layer) layer to encrypt the data. This requires minimal change in our code and is a widely used technique.

Exercises

- Experiment with a class to see what changes do or don't modify the generated serial version UID?
- Write a loop that creates a large number of simple objects (a million or more) serializes them one by one. Note the time taken (use `java.lang.System.currentTimeMillis()`). Then write another loop that creates the same objects but now place them into a `Collection` like `ArrayList`, `List` or `HashMap`. Now serialize the collection instead of serializing the objects one by one. Note the time again. Is there a significant difference? Is so, why?
- Research how to use the Kryo serialization framework for Java. Convert the program from the previous exercise to use *Kryo* and then time the program again.
- Research how to use the Protocol Buffers serialization framework for Java. Convert the program from the previous exercise to use *Protocol Buffers* and then time the program again.

References

- Other object serialization frameworks: Kryo, JSON, Google Protocol Buffers, Apache Avro, Facebook Thrift etc.
- Serialization (Wikipedia)
- Performance comparison of various serialization frameworks
- More on warnings when reading back a serialized object